Sky Beacon | One-Page Product Brief

A camera-first outdoor marker app for placing visible, GPS-backed beacons in the real world.

Audience: project partner / investor conversation | Perspective: user-first launch baseline | Date: June 16, 2026

Core idea: A user opens Sky Beacon outdoors, looks through their phone camera, chooses a beacon color, and places a luminous marker about 100 meters ahead. Later, when they turn back toward that saved location, the beacon reappears in the camera view as a visible directional cue.

User Experience

- No map-first setup: the real world stays as the main interface, with a reticle, live camera feed, and compact instrument-style controls.
- Placement is intentional: the user previews a full beacon, chooses from five colors, then confirms or cancels before anything is saved.
- The MVP is honest about accuracy: GPS and compass quality are shown clearly, and low-confidence placement is allowed but qualified.

Built Baseline

- Installable Next.js/TypeScript PWA with app manifest, icons, service worker, cached shell, and mobile-first camera experience.
- Progressive camera, GPS, and compass permission flows with sensor status, heading, stability, confidence, and HTTPS guidance for phones.
- Geospatial placement logic that calculates a beacon coordinate 100 meters ahead, then renders saved beacons by bearing through a DOM/CSS camera overlay.
- Beacon management drawer for up to three local beacons: select, rename, recolor, delete with undo, replace at limit, and clear all with confirmation.
- Browser-local persistence only; no account, backend, database, or cloud sync required for the first launch/demo.

Launch Outcome

Desired launch result: A polished MVP that someone can install or open on a modern phone, understand within one minute, and use to place a beacon within roughly 10 seconds after permissions are granted. The launch should prove the product feeling: aim, place, look back, and see the marker in the correct general direction.

After launch, the next outcome is precision: use field feedback to decide how Sky Beacon should evolve beyond approximate 100-meter placement into map/geospatial confirmation, optional map adjustment, shared/cloud-backed beacons, and eventually WebXR or visual refinement where the platform can support it reliably.